## 3.4 MAGNETIC PROPERTIES

All materials are not magnetic in nature. Further, all the magnetic materials will not behave identically. So, in order to differentiate one magnetic material from another, some basic parameters are used. They are:

**(a) Magnetising field**

**The magnetic field which is used to magnetize a sample or specimen is called the magnetising field.** Magnetising field is a vector quantity and is denoted by $\vec{H}$ and its unit is A $m^{-1}$.

**(b) Magnetic permeability**

**The magnetic permeability is the measure of ability of the material to allow the passage of magnetic field lines through it or measure of the capacity of the substance to take magnetisation or the degree of penetration of magnetic field through the substance.**

In free space, the permeability (or absolute permeability) is denoted by $\mu_0$ and for any other medium it is denoted by $\mu$. **The relative permeability $\mu_r$ is defined as the ratio between absolute permeability of the medium to the permeability of free space.**

$$\mu_r = \frac{\mu}{\mu_\circ} \qquad (3.29)$$

Relative permeability is a dimensionless number and has no units. For free space (air or vacuum), the relative permeability is unity i.e., $\mu_r = 1$.

**(c) Intensity of magnetisation**

Any bulk material (any object of finite size) contains a large number of atoms. Each atom consists of electrons which undergo orbital motion. Due to orbital motion, electron has magnetic moment which is a vector quantity. In general, these magnetic moments orient randomly, therefore, the net magnetic moment is zero per unit volume of the material.

When such a material is kept in an external magnetic field, atomic dipoles are induced and hence, they will try to align partially or fully along the direction of external field. **The net magnetic moment per unit volume of the material** is known **as intensity of magnetisation.** It is a vector quantity. Mathematically,

$$\vec{M} = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{\vec{p}_m}{V} \qquad (3.30)$$

The SI unit of intensity of magnetisation is ampere metre$^{-1}$. For a bar magnet of pole strength $q_m$, length $2l$ and area of cross-section $A$, the magnetic moment of the bar magnet is $\vec{p}_m = q_m \vec{2l}$ and volume of the bar magnet is $V = A\left|\vec{2l}\right| = 2lA$. The intensity of magnetisation for a bar magnet is

$$\vec{M} = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{q_m \vec{2l}}{2lA} \qquad (3.31)$$

In magnitude, equation (3.31) is

$$\left|\vec{M}\right| = M = \frac{q_m \times 2l}{2l \times A} \Rightarrow M = \frac{q_m}{A}$$



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